ADVANCED PROGRAMMING FOR BIG DATA

Home Credit Default Assessment

From imbalanced applicant data to a calibrated, cost-aware decision system

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PRESENTATION MAP:

- A1 Data
- A2 Select
- A3 Classical
- A4 Ensembles
- A5 Neural
- A6 Evaluate
- A7 Explain
- A8 Deploy



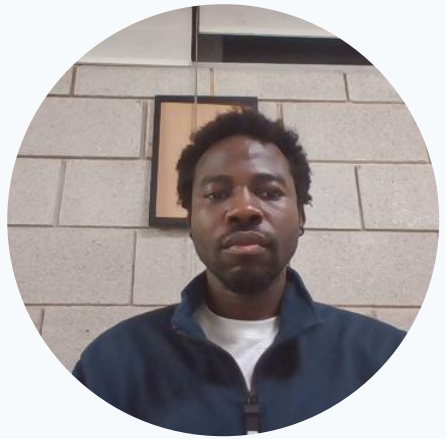
Recommendation first: calibrated LightGBM, not an automatic lender

The rest of the deck earns this recommendation section by section.

0.7673	0.2603	0.0031	0.1613	4,277
HELD-OUT ROC-AUC	HELD-OUT PR-AUC	CALIBRATED ECE	THRESHOLD	SAVING
XGBoost: 0.7667	baseline: 0.0807	from 0.3167	OOF cost choice	cost units

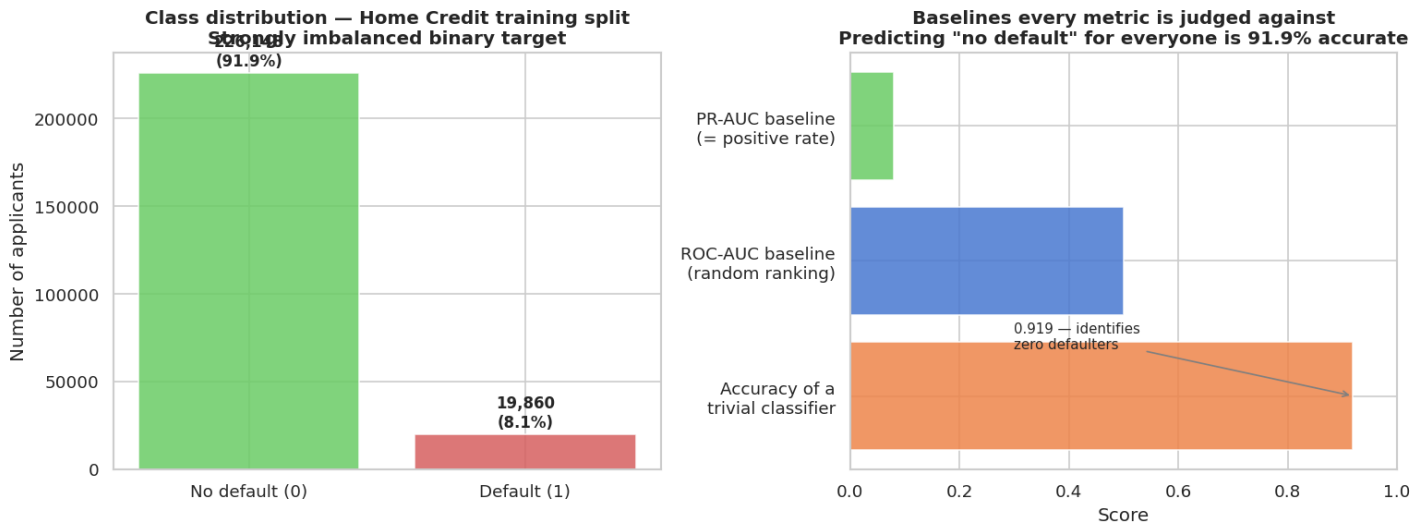
I recommend LightGBM with Platt scaling and a 0.1613 operating threshold as monitored decision support. It ranks first in development, stays marginally ahead on the sealed test set, and becomes usable only after calibration and threshold checks.

Not recommended: for autonomous approval or rejection. The model should support a risk team, with appeal, monitoring and revalidation before live use.



Only 8.07% default, so accuracy is the wrong headline

A trivial “no default” model looks accurate but catches nobody.



ROW CHECK: No applicant rows were dropped for missing values; missingness is handled by train-fitted imputation and indicators.

DECISION

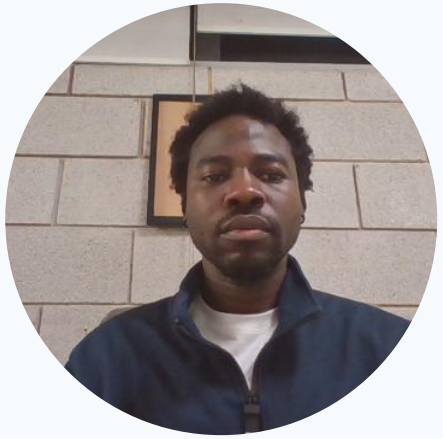
Use ROC-AUC and PR-AUC first.

EVIDENCE

307,511 labelled applicants: 246,008 development rows and 61,503 sealed-test rows.

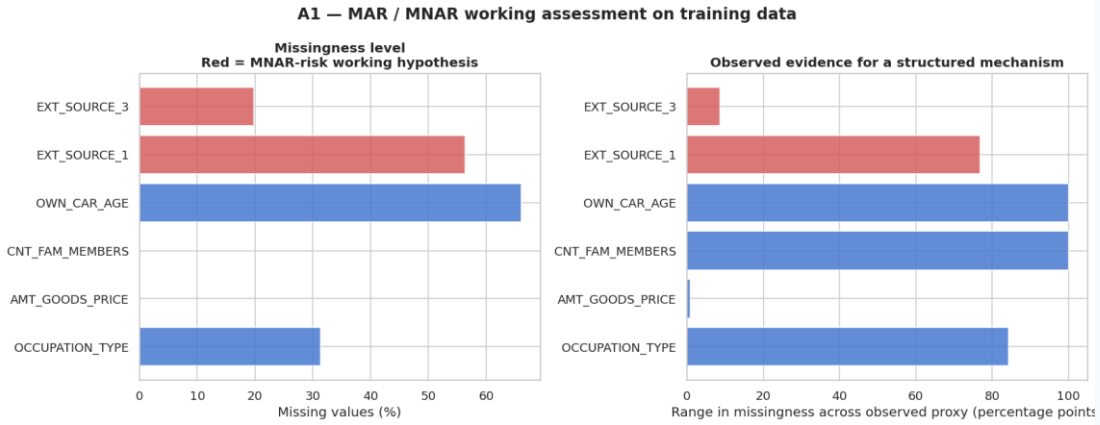
WHY IT MATTERS

Accuracy would reward the majority class and hide failure on defaulters.



Missingness has meaning, so I did not use one blanket imputation

The MAR/MNAR labels are working hypotheses based on training-only evidence.



Feature	Call	Evidence used
OCCUPATION_TYPE	MAR	income type varies
AMT_GOODS_PRICE	MAR	contract type
OWN_CAR_AGE	structural	car ownership
EXT_SOURCE_1	MNAR-risk	external data footprint
EXT_SOURCE_3	MNAR-risk	education patterns

Decision: fit imputers inside the pipeline and keep missing indicators. Otherwise, absence of data would be silently converted into fake average applicants.

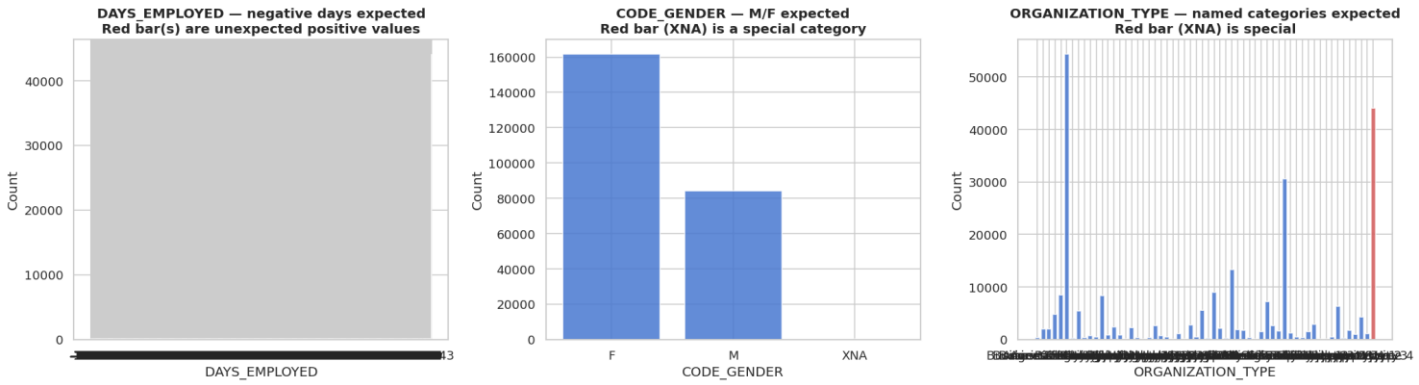
MISSING DATA ROW CHECK: 67/122 columns contain missing values; 7.3M cells are missing, but 0 rows are removed only for missingness.

ROWS LEFT: 307,511 labelled applicants remain: 246,008 development + 61,503 sealed test.



365,243 employment days is a sentinel, not real tenure

The exact co-occurrence with ORGANIZATION_TYPE = XNA makes it a structural signal.



DECISION

Replace the sentinel with NaN and add a flag.

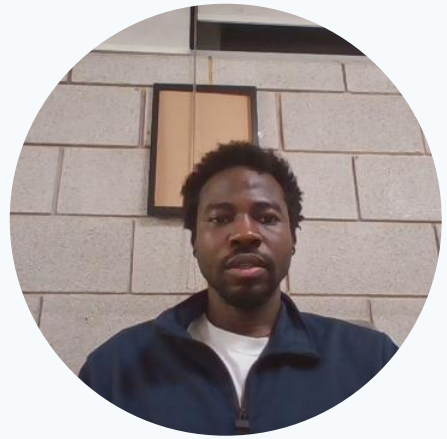
EVIDENCE

44,143 rows carry 365,243; the same rows have ORG_TYPE = XNA.

WHY IT MATTERS

Leaving it numeric would create a fake 1,000-year tenure pattern.

ANOMALY ROW CHECK: 44,143 sentinel rows are recoded, not deleted; labelled rows remain 307,511.



Applicant history was aggregated before joining to avoid duplication

This is the Merit/Distinction lever: one applicant stays one row.

bureau
1.7M rows

→

bureau_balance
27.3M rows

→

previous
1.67M rows

→

installments
0.81M rows

→

POS/CASH
10.0M rows

246,008 → 246,008
train rows kept after joins
lost rows: 0 · duplicate rows: 0

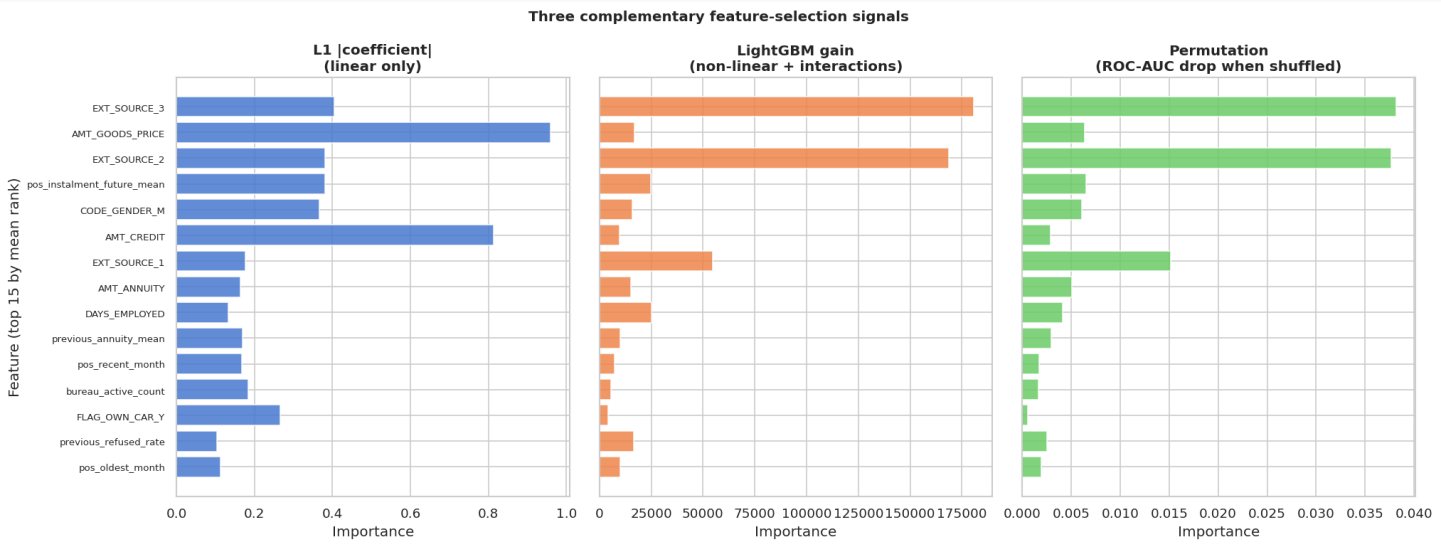
Feature family	Example	Why it was built
Affordability	credit / income	payment burden
Credit history	bureau debt mean	past exposure
Repayment	instalment future mean	payment pressure
No-history flags	no bureau history	absence is informative

If raw one-to-many tables were joined directly, applicants would repeat and the target relationship would be corrupted.



Tree-based selection agreed most with permutation evidence

No single importance method was trusted on its own.



ρ = 0.677

LGB VS PERMUTATION

12/15 overlap

21

FINAL SUBSET

source columns

Decision: use the union of top-15 features across L1, LightGBM gain and permutation importance. The reduced set carries the nonlinear signal forward.



The model became about three times faster for only 0.010 AUC

The smaller feature set was good enough to carry into A3 to A8.

287
FULL WIDTH
model columns

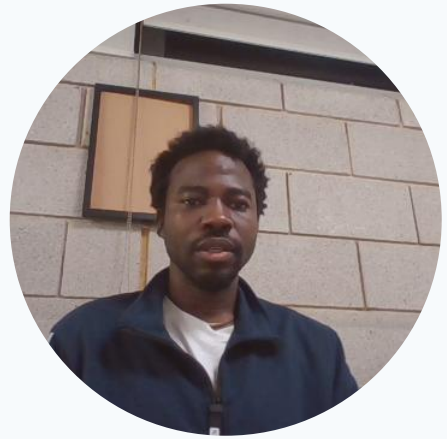
98
REDUCED WIDTH
model columns

33.3s
TIME SAVED
49.1s to 15.8s

-0.0101
AUC CHANGE
acceptable cost

Set	Columns	ROC-AUC	Fit time
Full	287	0.7702	49.1s
Reduced	98	0.7600	15.8s
Change	-189	-0.0101	-33.3s

What would go wrong: keeping all transformed columns would slow every later search and increase redundant noise without a large gain.



Every model sees preprocessing fitted inside the training fold only

This is the point where EDA choices become a leakage-safe modelling pipeline.

21 source features → **split by column type** → **impute + log/scale/OHE** → **fit on training fold** → **score untouched validation fold**

Column type	Treatment
Skewed numeric	log1p + median + scale
Other numeric	median + standardise
Categorical	impute + one-hot
Sealed test	used only at final check

If preprocessing was fitted before cross-validation, validation data would influence transformations and inflate the model comparison.



Linear models plateau near 0.747 AUC; k-NN is the tuning exception

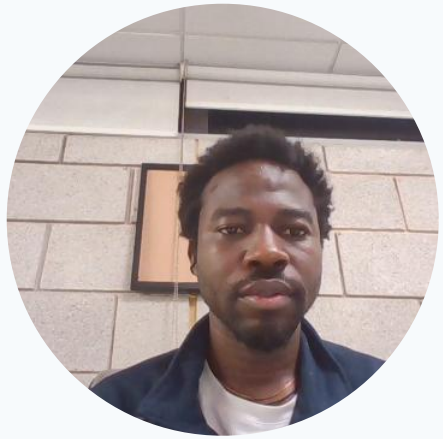
All four classical families used the same reduced feature set and folds.

Model	Base	Tuned	Gain	Main setting
Logistic	0.7465	0.7466	+0.0001	C=0.01
Linear SVM	0.7465	0.7466	+0.0000	C=0.01
k-NN	0.6945	0.7263	+0.0317	k=75
Naive Bayes	0.6384	0.6384	+0.0001	smoothing

0.7466
BEST CLASSICAL
Logistic / SVM

+0.0317
K-NN GAIN
tuning helped

Decision: keep the simple linear classical baseline. Extra tuning barely moved the linear methods.



Classical families were limited more by model form than by tuning

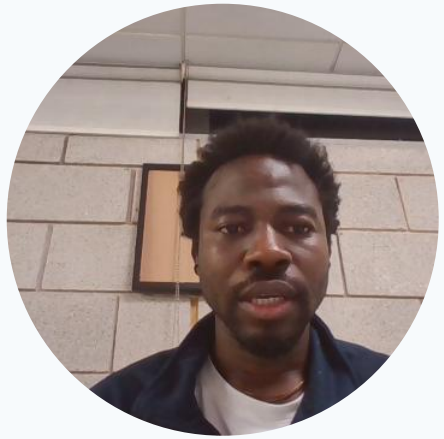
The results match the theory: sparse tabular data rewards ranking, not distance voting.

Family	What happened	Reading
Logistic	stable at 0.7466	linear boundary was enough for a baseline
Linear SVM	same ranking ceiling	margin model behaved like logistic
k-NN	largest tuning gain	distance struggled in sparse one-hot space
Naive Bayes	lowest AUC	independence assumption too strong

0.6805
LINEAR SVM PROBE
small 5k test

0.6858
RBF SVM PROBE
13.3s vs 0.5s

Consequence: a costly kernel would slow the workflow without enough measured improvement to justify it.



Boosting wins the ensemble comparison, with LightGBM just ahead

Random Forest improved with tuning, but boosting captured more signal.

Model	Base	Tuned	Gain	Search time
Random Forest	0.7434	0.7487	+0.0053	736s
XGBoost	0.7606	0.7618	+0.0012	356s
LightGBM	0.7617	0.7621	+0.0004	331s

0.7621
CV LEADER
LightGBM

-0.0003
XGBOOST GAP
practical tie

Decision: carry LightGBM as the development winner, while treating XGBoost as a serious alternative rather than a failed model.



Boosting fits the interaction-rich signal better than bagging here

The A2 tree/permutation agreement already suggested nonlinear structure.

Random Forest: independent deep trees

Tree 1 → Tree 2 → Tree 3 → Average

0.7487 AUC

XGBoost / LightGBM: trees correct residual error

Tree 1 → Tree 2 → Tree 3 → Tree 4

0.7621 AUC

Consequence: if the model family cannot capture interactions between affordability, credit history and external scores, tuning alone cannot close the gap.



The same MLP was implemented in Keras and PyTorch

Architecture, folds and reduced inputs were held as constant as possible.



0.001
LEARNING RATE
selected

64
WIDTH
selected

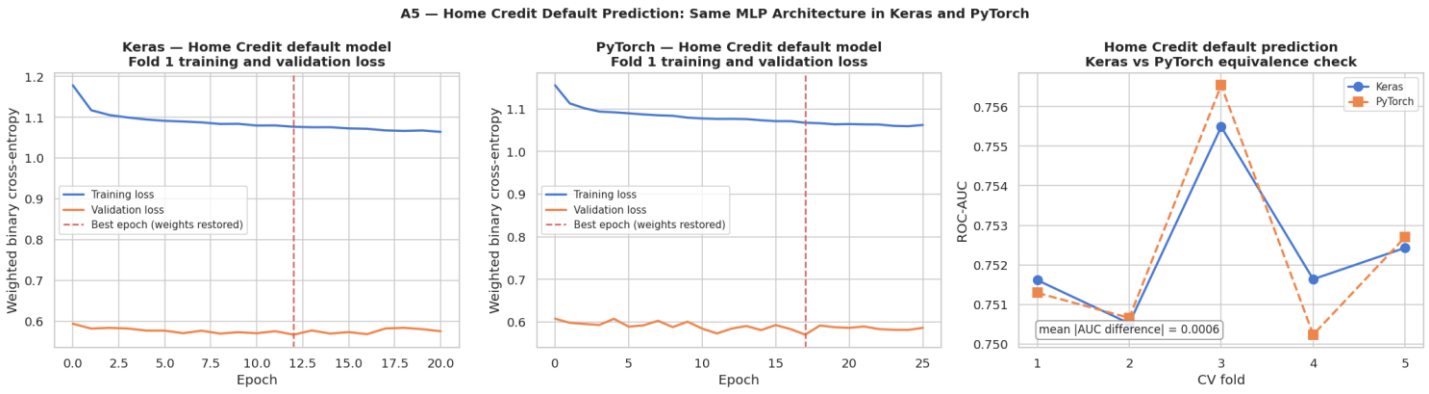
0.20
DROPOUT
selected

Decision: use Keras once in the formal leaderboard and keep PyTorch as a replication check, not a separate model family.



Keras and PyTorch agree closely enough for a replication check

The second implementation supports consistency, not a new family in the leaderboard.



0.7523

KERAS CV AUC

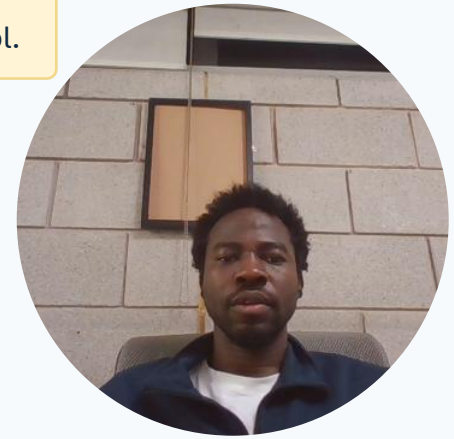
tuned MLP

0.7523

PYTORCH CV AUC

same architecture

Production reflection: Keras is quicker for this tabular baseline; PyTorch is better when the training loop needs more control.



LightGBM leads, but XGBoost is practically tied

A single leaderboard keeps model order and metrics consistent.

Model	ROC-AUC	PR-AUC	Recall	Brier ↓
LightGBM	0.762	0.249	0.672	0.187
XGBoost	0.762	0.248	0.667	0.186
MLP Keras	0.752	0.234	0.675	0.191
Random Forest	0.749	0.226	0.616	0.187
SVM linear	0.747	0.223	0.011	0.069
Logistic	0.747	0.222	0.677	0.204
k-NN	0.726	0.207	0.001	0.069
Naive Bayes	0.638	0.123	0.604	0.324

0.0807

PR BASELINE

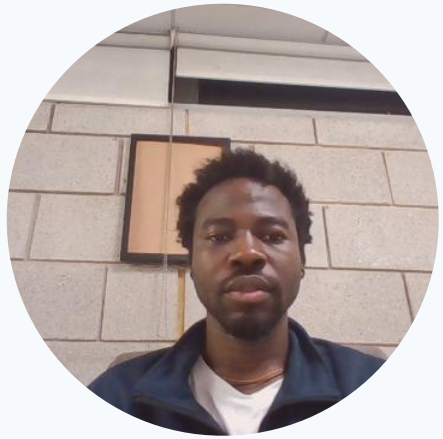
default rate

0.0003

TOP AUC GAP

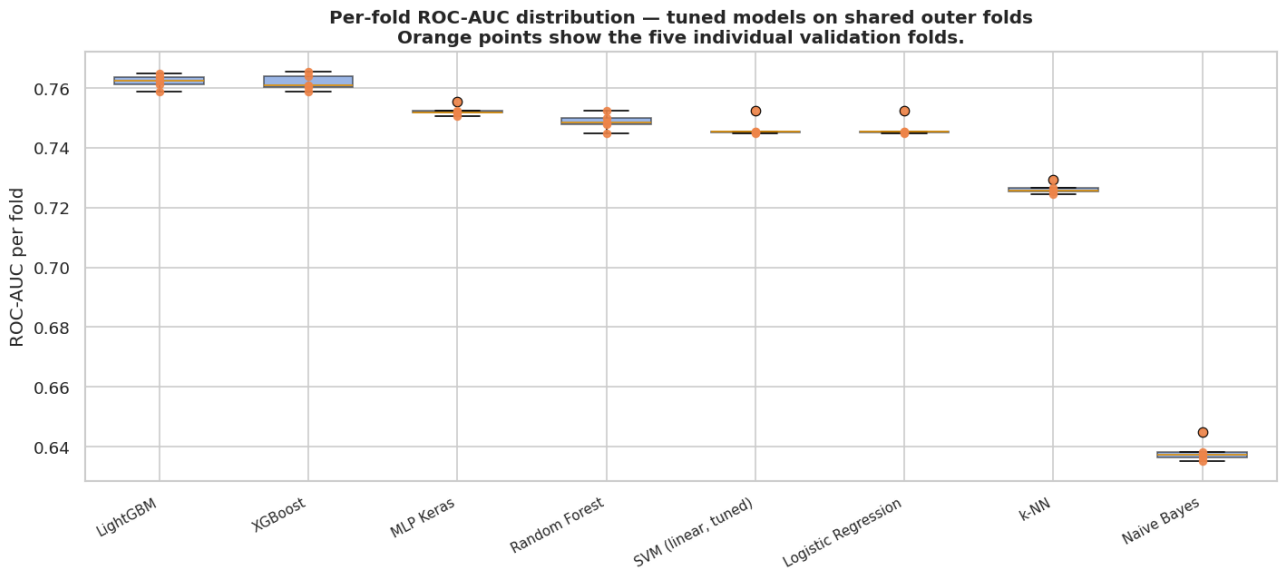
LGBM vs XGB

Consequence: the winner is selected from development evidence, not by retuning around the sealed test set.



Overall differences are real; five folds limit pairwise claims

Magnitude, direction and fold consistency carry the practical argument.



0.00003
FRIEDMAN P
reject equal ranks

0.00179
BONFERRONI A
28 tests

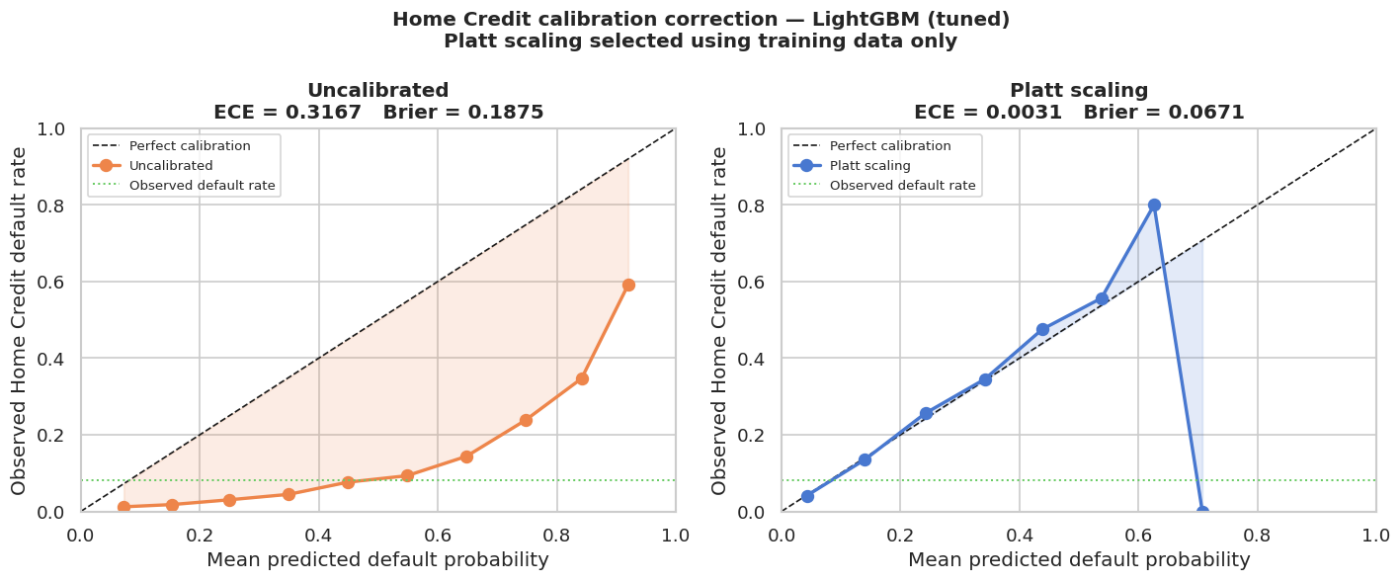
0.0625
MIN EXACT P
only five folds

Pairwise significance is mathematically unreachable at this resolution after Bonferroni, so the box plot is essential evidence.



Platt scaling cuts LightGBM ECE from 0.3167 to 0.0031

Ranking remains the same; the probability scale becomes usable for decisions.



0.3167
RAW ECE
class-weighted scores

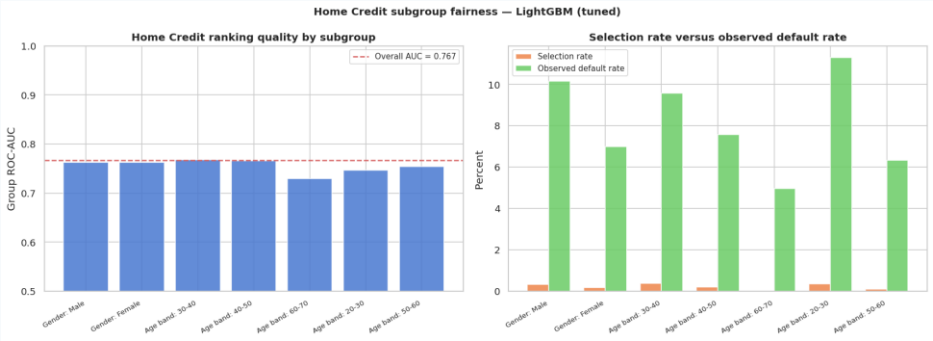
0.0031
PLATT ECE
held-out test

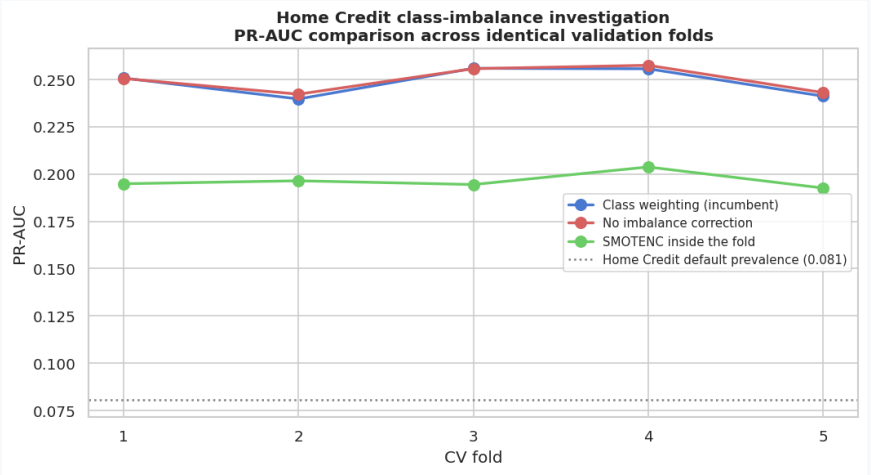
0.0814
MEAN P
actual: 0.0807



Age is the main subgroup risk; SMOTENC made performance worse

Fairness is a monitoring condition, not a label earned from one chart.





0.039

AGE AUC RANGE

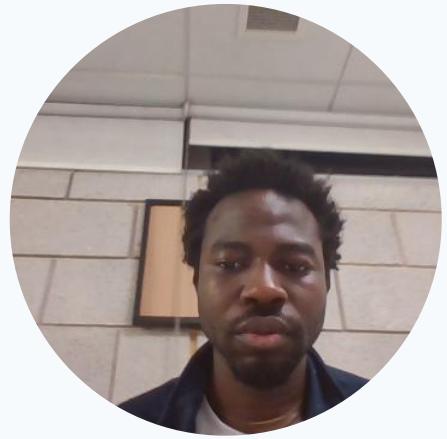
weakest: 60–70

-0.052

SMOTENC PR CHANGE

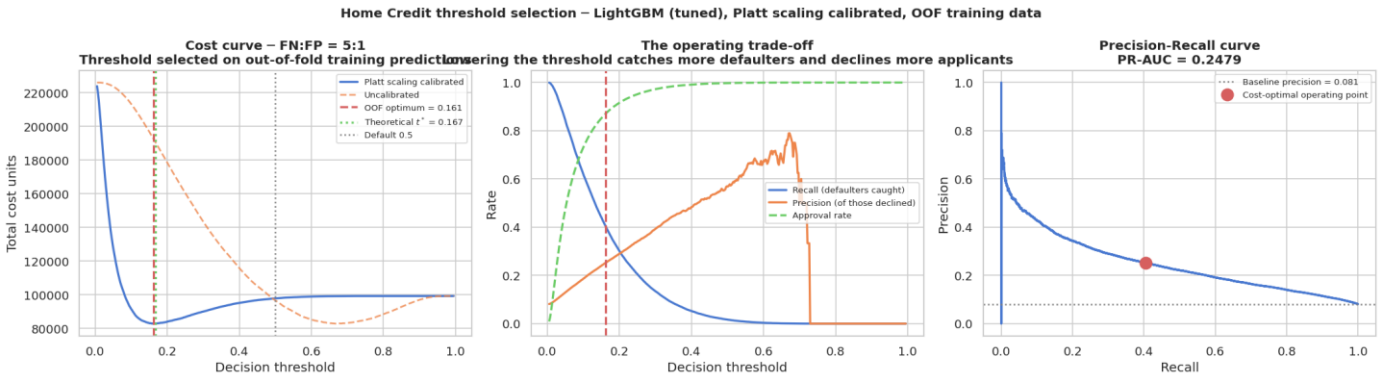
vs class weight

Decision: keep class weighting and calibrated probabilities. Do not deploy synthetic-applicant training here without a clear measured benefit.



A 0.1613 threshold saves 4,277 assumed cost units

The cost ratio is an assumption, so it must be revalidated before live use.



43%

RECALL

defaulters caught

26%

PRECISION

among declined

87%

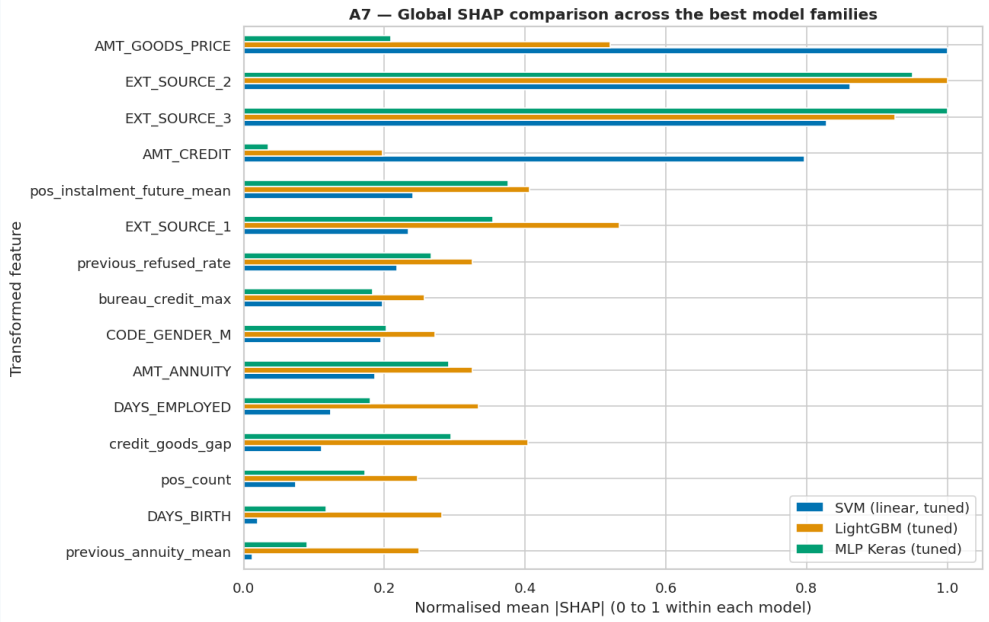
APPROVAL RATE

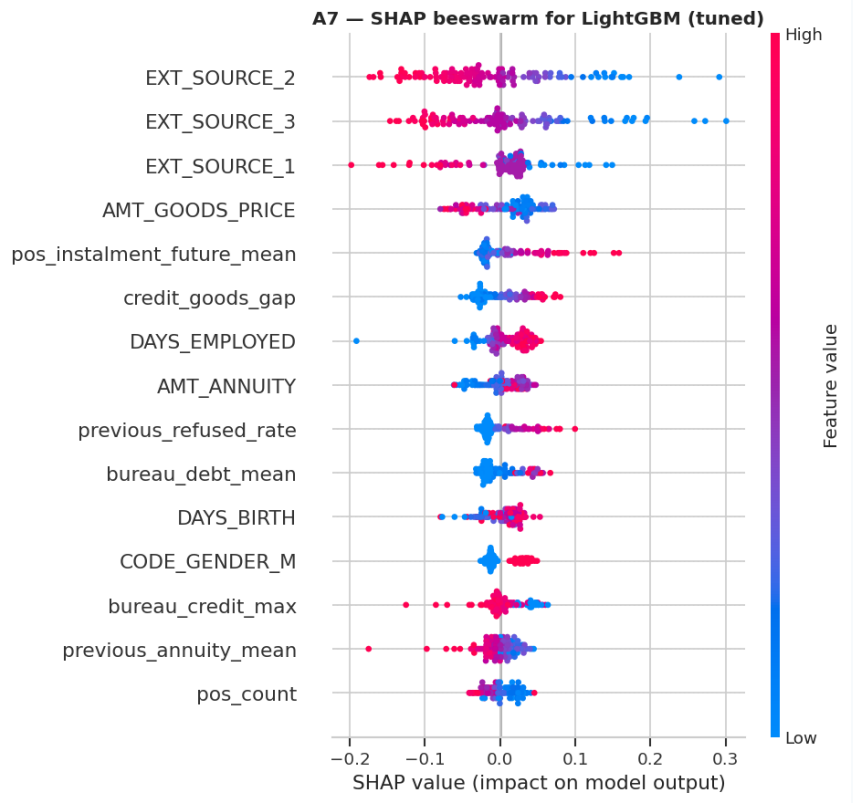
test applicants



SHAP agrees that external scores and affordability dominate risk ranking

The same applicants and transformed columns are compared across model families.



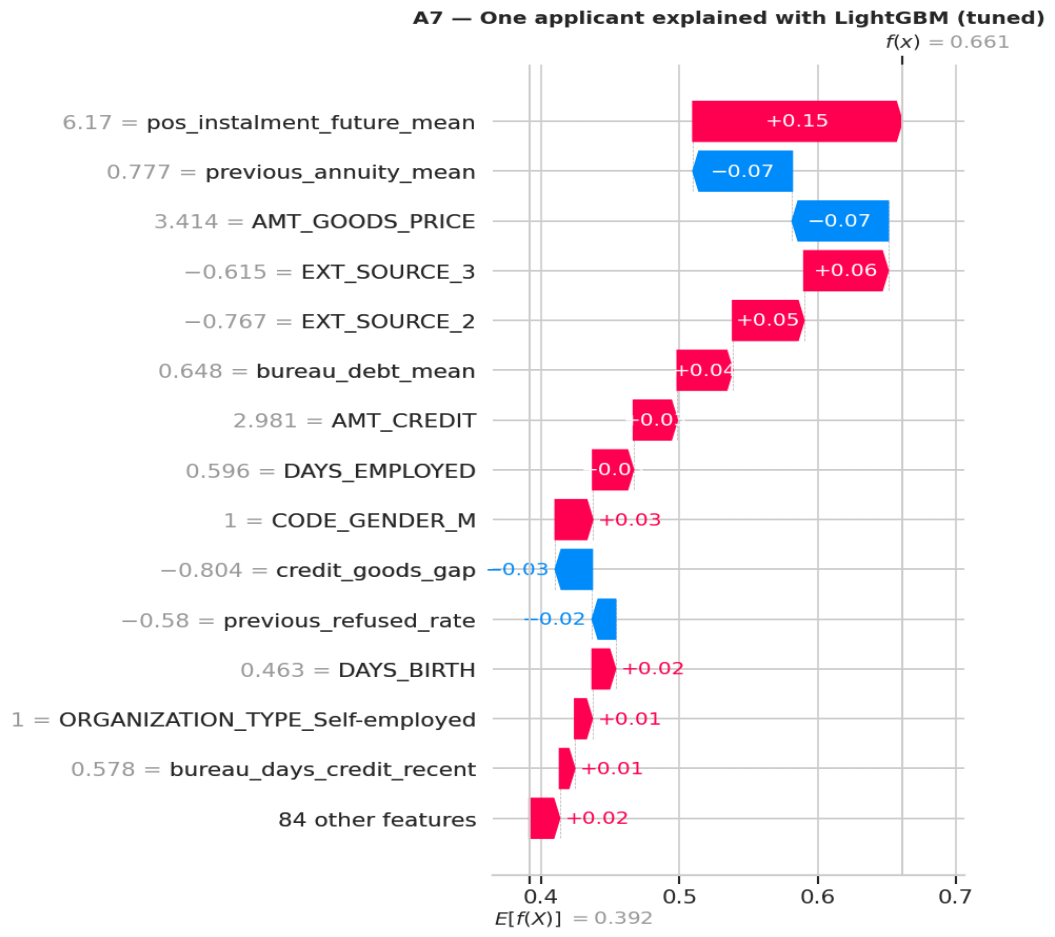


Reading: SHAP is model behaviour, not causality. It explains why the fitted system scores an applicant as risky, not how to change real-world risk.



The false positive sits only 0.005 above the decision threshold

This is exactly why human review still matters near the boundary.



No default

ACTUAL OUTCOME

TARGET = 0

0.1664

MODEL PROBABILITY

threshold: 0.1613

0.0051

FLIP DISTANCE

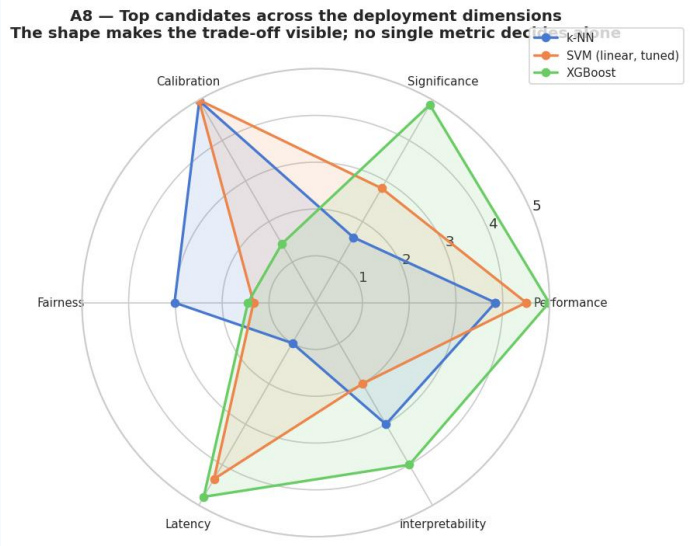
probability points

Largest upward pushes: future instalment burden plus low EXT_SOURCE scores. A tiny downward probability move would cross the threshold.



Deploy only as controlled decision support, not autonomous lending

The final recommendation is a decision system, not just a high AUC score.



Question	Answer
Best development model?	LightGBM
Statistically secure alone?	No: low pairwise power
Probability usable?	Yes, after Platt scaling
Fair enough to deploy?	Only with monitoring
Operationally acceptable?	Yes: 4.46 ms / 1k

Model card conditions: revalidate the 5:1 cost ratio, monitor calibration and subgroup gaps, keep appeal/human review, and version the pipeline, calibrator and threshold together.

Close: historical repayment data can reproduce old policy and access patterns; a better model alone cannot remove that limitation.

